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A Pre-Trained Model for Aspect-based Sentiment Analysis Task: using Online Social Networking

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Abstract

Aspect-based sentiment analysis (ABSA) is a crucial part of Natural Language Processing (NLP) that focuses on identifying emotions related to specific elements in written material. ABSA has gained widespread interest due to its ability to provide precise insights into sentiment expressions across different domains. Social media provides a valuable resource for ABSA, containing user-created content with viewpoints and feedback. However, the informal nature of social media text poses challenges for ABSA. This study investigates the performance enhancement of baseline and proposed models in the ABSA task context. Both baseline and proposed models were evaluated for accuracy and F1 score improvements. The results showed that the suggested model performs better than other baseline models, with an improvement of 0.52% F1 score and an overall accuracy increase of 1.16%. However, specific analyses for laptops indicated limitations in the proposed model's performance, with accuracy scores ranging from 72.65% to 84.98%.

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Keywords: ABSA,BERT,SentencePairClassifcation,SemEval

1. Introduction

In recent years, social networking has completely changed how people communicate. It was first utilised as a crucial online tool for maintaining contact with friends and acquaintances. Still, it has since developed into intricate systems that influence our relationships, change our beliefs, and transform communication [1]. Despite geographical restrictions, social networking's capacity to link people and communities globally is its fundamental draw. However, it is crucial to comprehend the enormous impacts of these platforms on society, psychology, business, and other areas as we traverse the constantly evolving digital world. This essay deeply dives into social networking, examining its development from basic Internet message boards to the sophisticated networks we use today. The writers also examine how social networking has become crucial to our daily lives due to technical developments and cultural shifts. Even though these platforms have many advantages, we must address problems like isolation, echo chambers, and dissemination of false information [2, 3]. As we advance into the digital era, it is crucial to understand the benefits and drawbacks of social networking and address these problems. Our goal is to present a thorough analysis of social networking's effects on society. We examine the democratisation of information and the amplifying of individual voices, the difficulties of establishing an online identity, and the fuzziness of lines separating the private and public spheres. We can traverse its complexities and harness its power for the greater good by

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clearly understanding its past, present, and potential future [4]. In the following pages, we examine the intricate threads that make up the fabric of social networking, bringing together technology, psychology, sociology, and more [5]. As we unpack its complexities, we encourage readers to reflect on their roles in this digital landscape and consider how these networks have impacted our lives and continue to shape our future. Figure 1 visually represents the Facebook users on online social media. The graphical depiction is informative and gives a clear picture of the Facebook user base. It is evident from the presented data that the age group of 25 to 34 years old represents the majority of users. In comparison to other age groups, this one has the most users.

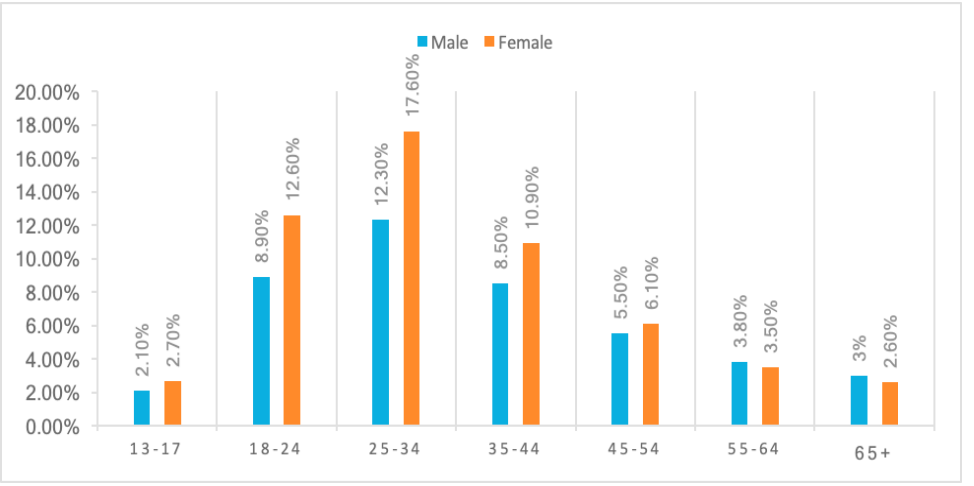


Fig. 1. Global Facebook members as of January 2023, divided by age and gender

We may learn much about the modern digital world using social networks, which make sharing thoughts, opinions, and feelings simple. These networks, combined with sentiment analysis technology, enable us to comprehend the feelings of humankind as a whole on a never-before-seen scale. With the aid of technology, we can explore the internet environment and interpret hidden sentiments in tweets, comments, and status updates [6]. Social media platforms in this age of digital interconnection provide us with a wealth of ideas, viewpoints, and feelings. But because of sentiment analysis technologies, we can now gather incredibly insightful knowledge about the overall emotional state of humanity on a never-before-seen scale. Thanks to technology, we can efficiently explore the internet environment and unravel the underlying sentiments behind status updates, tweets, and comments thanks to technology. In today's digital age, many text data sources exist, including social media, product reviews, news stories, and more. This information can provide important new perspectives on how people feel and behave. Sentiment analysis has become a potent technique for gaining these insights as technology becomes more pervasive in our daily lives. Sentiment analysis enables us to interpret textual content's feelings, viewpoints, and finer details [7].

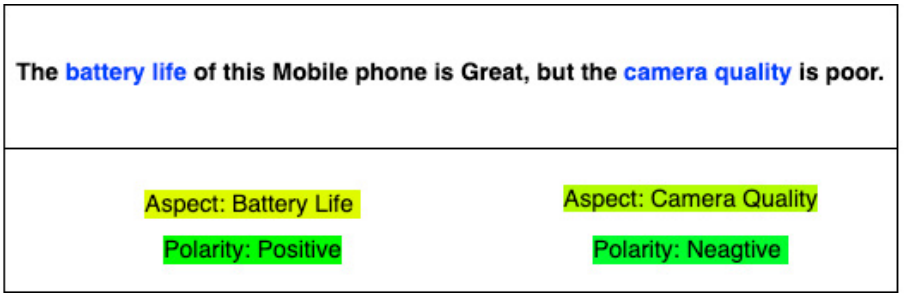


Fig. 2. Exemplary Sentence of ABSA

In this paper, authors have used ABSA, computational linguistics, machine learning(ML), and NLP. This method automates deciphering verbally spoken human emotions, opinions, and attitudes. Whether a text expresses a good, negative, or neutral sentiment, sentiment analysis examines that emotion. This aids in determining consumer happiness, brand impression, public opinion / Reviews, and even the forecasting of market trends. Figure 2 shows the review statement of ABSA, where in one statement, two Aspects are presented of different polarity. ABSA has developed into a useful tool across numerous disciplines, including business, marketing, politics, healthcare, and social sciences, in today's age of information overload, where billions of text-based exchanges occur daily [6]. Organisations may make data-informed decisions, customise their communication strategy, and obtain deeper insights into consumer behaviour by revealing the sentiment buried within the lines of text. Improving BERT-SPC (Sentence-Pair Classification) models for ABSA tasks is crucial for several reasons, including accuracy, adaptability, comprehensive sentiment analysis, and eliminating bias issues. These improvements can have a significant impact on various industries and user experiences, reinforcing the importance and value of ABSA in today's data-driven world.

2. Related Work

[8] To determine user sentiment orientation on online social networks, the researchers developed a novel USON model that blends user interaction and opinions. They analyzed the sentiment polarity distribution of the opinion datasets to determine the user sentiment orientation score. They also developed a user social network that uses node attributes to display user sentiment ratings and edges to show social connections. To analyse sentiment, they merged user and opinion representations. The model was tested on Yelp and Amazon, which revealed that it outperformed other current methods.[9] This paper proposes a new and novel approach for absa called the FOL-integrated logic tensor network with massive rules (LTNMR). The method incorporates two types of knowledge into the network: dependency knowledge to help identify aspect-related words and human-defined knowledge rules to guide the sentiment analysis of those words. The authors also introduce a technique called mutual distillation structure knowledge injection (MDSKI) to improve the accuracy of the inference. Bert, the instructor, teaches the dependency knowledge to LTNMR, the student network, using MDSKI. The experiments show that LTNMR, when paired with MDSKI, outperforms other state-of-the-art methods for ABSA. [10] This article explores how stakeholders describe their positive and negative experiences with health and care integration in care homes and whether these experiences changed during the pandemic. The authors used network analysis on post-estimation data to determine the emerging main themes. Before the pandemic, evaluations that included integration terms were generally more positive than those that did not. This trend continued throughout the first year of the pandemic. During the COVID-19 period, there was an overall increase in positive sentiment (+1.1 points on average). This was mainly due to reviews that mentioned integration phrases at the beginning of the first and second waves. [11] In this study, the authors utilised multilabel learning techniques to create a gender-based sentiment detection model that considers multiple aspects. They divided the Abilify and Celebrex datasets into three groups based on gender: mixed, male, and female. Then, they analysed each group's characteristics using bag-of-words (BoW), term frequency-inverse document frequency (TF-IDF), and global vectors for word representation (GloVe) data. They implemented multichannel recurrent neural networks with attention mechanisms and transformation techniques to further enhance their model. The results revealed that deep learning models using GloVe word embedding features performed better regarding mean test accuracy, AUC Score, RL, and average precision in both datasets. However, they found that traditional multilabel transformation methods were more effective for small amounts of data and long-range sequences regarding samples and labels. [12]This study presents a new model called the Dependency Tree Graph Convolutional Network (DTGCN) for sentiment classification. The model utilises Chinese syntactic dependency trees and graph convolution to create syntax-dependent trees and learn contextual features using the Bi-GRU model. These trees' adjacency matrices are then used to obtain sentiment features with GCNs. These features are fed into the SoftMax classifier to determine the sentiment polarity. The findings indicate that the DTGCN model performs 4.45% better than benchmark models, with 90.51% accuracy and 90.34% recall rate. This demonstrates how the model can use subtle grammatical differences in Chinese text to enhance sentiment categorisation. The suggested model also offers a novel method for discovering public opinion,

improving sentiment categorisation, and finding hidden links in expressions. [13]This study examines how user-generated information, particularly online evaluations, can forecast a restaurant's survival by analysing data from eateries in two well-known tourist areas in the United States. Location, taste, pricing, service, and atmosphere make up the five categories that ABSA uses to classify the overall review attitude. The study makes use of a conditional survival forest model that is based on machine learning. It proves that, in comparison to overall review sentiment, Absa for several criteria can increase the accuracy of restaurant survival forecasts. The research also used feature significance analysis to discover which components of Internet evaluations are the best indicators of restaurant survival. It emphasises the influence of several types of aspect sentiment on the prognosis of restaurant survival. [14]The authors of this article introduced the GSAAN-BMA-SA-OPR method, a fresh approach to creating online product suggestions. Information from the Amazon product database is gathered using a Graph Sample, Aggregate Attention Network (GSAAN), Barnacles Mating Algorithm (BMA), and sentiment analysis (SA). Using GSAAN and BMA, the recommendations are rated as exceptional, good, very good, dreadful, or extremely bad. The authors used the accuracy, F-Score, recall, and precision measures to assess the model's performance once it had been built using MATLAB. In comparison to existing models, our suggested method produces greater Mean Absolute Percentage Error (MAPE) (23.77%, 65.3%, and 10.42%) and lower Mean Squared Error (MSE) (42.15%, 29.64%, and 32.48%). [15]In their study, the authors provide a novel flood forecasting and monitoring model that uses a transformer network to assess the significance of a flooding occurrence through sentiment analysis of the multimodal inputs (text and images). The authors also provide an experimental evaluation of many state-of-the-art deep learning methods for image and natural language processing. Finally, they demonstrate how data collected from tweets may be used to dynamically and instantly present data related to fine-grained geographic flooding.

3. Proposed work

The authors have opted for this study's BERT Sentence Pair Classification (SPC) [16] model. This model is beneficial for problems that involve pairs of sentences. Using Bert's CONTEXTUAL LANGUAGE KNOWLEDGE, the BERT-SPC model can classify pairs such as questions, responses, premises, and hypotheses. The input sentences are fed into BERT's layers to achieve this, allowing the model to understand complex contexts. The sentence embedding is combined and further processed with fully connected layers and pooling, making categorisation easier. This unique architecture makes the model well-suited for sentiment analysis and natural language inference tasks, as it can effectively interpret the relationships between phrase pairs.

Figure 3 illustrates the proposed work of this study, where authors have used the BERT-SPC Pre-trained model for the ABSA task. The study recommends utilising the BERT-SPC pre-trained model for the ABSA task, as shown in Figure 3. The first step is to pre-process the SemEval 2014 dataset to use it effectively, and authors should get the best results.

3.1. Pre-processing

Before text can be used for activities such as analysis, NLP, and ML, it must undergo pre-processing [17]. This involves modifying, cleaning, and organising unstructured text into a format that algorithms can effectively use. Figure 4 outlines the standard process for preparing the text, including various steps such as cleaning the text, tokenising it, removing stop words, eliminating URLs and HTML tags, and removing non-alphanumeric tokens. To remove the Url, they are been replaced with space. There are many NLP libraries available, such as spaCy and NLTK, that provide functions for text tokenization and preprocessing. These libraries can handle non-alphanumeric tokens and other preprocessing tasks. In text cleaning, symbols, punctuation, and special characters are removed. All text is converted to lowercase to ensure consistency. Abbreviations that expand or contract are also properly handled. Tokenisation is a process that involves breaking down a document into individual words or tokens and also breaking down sentences into their component words, which can be analysed [18].

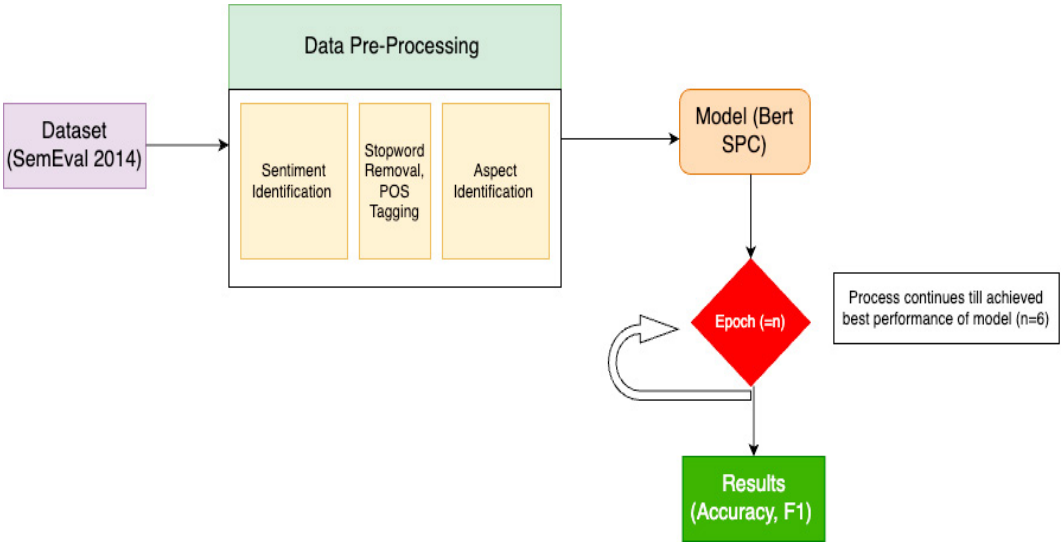


Fig. 3. BERT- SPC Model for ABSA task.

3.1.1. Stop- Word Removal

Stop words are words which is considered a value in text analysis because they are very common and do not typically carry much meaning on their own (e.g., ”the,” ”and,” ”in,” and ”is”). Here’s how stop words are typically removed in the preprocessing steps:

1. Tokenization is the initial step in text preprocessing. It involves breaking the text into individual words or tokens. The text is split into a list of words using whitespace and punctuation as delimiters.
2. A stop word list is a predetermined collection of words that are commonly excluded from natural language processing (NLP) tasks. These words are typically common words such as ”a,” ”an,” ”the,” ”in,” ”on,” and so on. Different stop-word lists are available for various languages and NLP libraries. You can also create your custom list of stop words if necessary.
3. Following tokenization, every token is checked against a list of stop words. If a token is found to match a stop word, it is eliminated from the list of tokens. Conversely, if a token does not match any stop word, it is kept in the list of tokens.
4. After removing stop words, additional text preprocessing steps like stemming or lemmatization can normalize and reduce data dimensionality.

3.2. BERT- SPC

BERT-SPC (Bidirectional Encoder Representations from Transformer with Sequential Pairwise Classification), a natural language processing model. ABSA involves determining and categorising the sentiment of a specific part of a sentence. BERT-SPC is a modified version of BERT, a pre-trained language model useful for various natural language processing applications. BERT-SPC differs from BERT in two ways [16]:

- The sentence, aspect, and separator token are concatenated to form the input sequence for BERT-SPC, enabling it to understand the relationship between the sentence and the aspect.

- A sequential pairwise classification layer is the BERT-SPC model’s final layer. Using the BERT encoder’s hidden states, this layer classifies the following token pairs into positive, negative, or neutral.

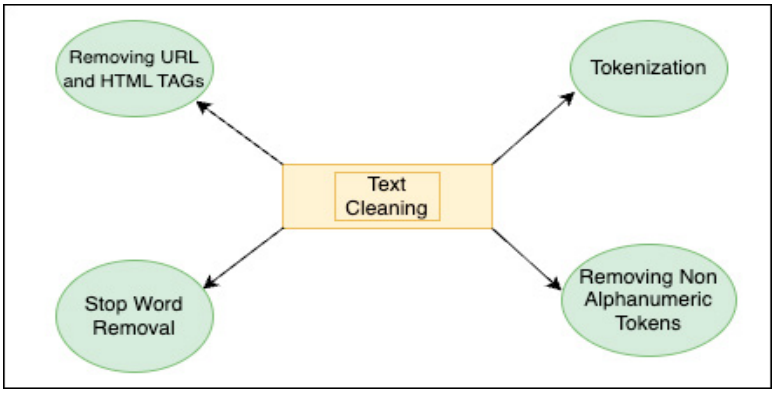


Fig. 4. Steps Used in Text Cleaning

This layer enables BERT-SPC to outperform BERT in ABSA tasks by learning the text's mood towards the aspect. BERT-SPC is a reliable model that may be used for various ABSA jobs because it has produced state-of-the-art results in multiple ABSA benchmarks [17, 18]. Utilising BERT-SPC has many advantages, including its comparative simplicity of training and capacity to give cutting-edge outcomes on numerous ABSA criteria. It is a strong and effective model that can potentially raise ABSA performance greatly.

4. Experimental Details

This section describes the experimental setup and conditions used to calculate the efficiency and logic of the BERT-SPC model in interpreting the experiment's data.

4.1. Dataset

The SemEval (Semantic Evaluation) workshop compares the effectiveness of several natural language processing (NLP) systems across diverse tasks. Researchers compare the effectiveness of their approaches using SemEval's standard datasets and evaluation metrics. One of SemEval's tasks, Task 4 [19], was focused on "Sentiment Analysis in Twitter" during the SemEval 2014 workshop. Its primary objective was to develop sentiment analysis methods that could identify the sentiment expressed in tweets and automatically classify them as positive, negative, or neutral despite the brief, informal nature of the material. Figure 5 shows the sample dataset used in this study.

Sample of dataset:

*"I loved their **fajitas**" → {fajitas: positive}*
*"I hated their **fajitas**, but their **salads** were great" → {fajitas: negative, salads: positive}*
*"The **fajitas** are their first plate" → {fajitas: neutral}*
*"The **fajitas** were great to taste, but not to see" → {fajitas: conflict}*

Fig. 5. Sample of Dataset.

Table 1 provides a detailed breakdown of instances for each domain dataset used in our study. The three domain datasets used were Laptop, Twitter, and Restaurant.

Table 1. Dataset Description

Dataset		Instances	Neutral	Positive	Negative
Laptop 14	**Train**	2,328	464	994	870
	Test	638	169	341	128
Restaurant 14	**Train**	3,608	637	2164	807
	Test	1,106	196	728	182
Twitter 14	**Train**	6,248	3,127	1,561	1,560
	Test	692	346	173	173

In our proposed study, Table 2 outlines the parameters that we have described. These parameters are crucial for achieving optimal results in the ABSA task and form the backbone of our research.

Table 2. Parameters used in the study

Used Parameter	Assigned Value	Description
BERT Model	Bert-Base-Uncased (12 layers 110M Parameter)	This is a pre-trained English language model designed for masked language modelling.
Batch Size	64	The batch size of Bert is 8, 16, 32,64 and 128.
Learning Rate	5e-5	The Learning rate is crucial for training deep learning models like BERT. It adjusts model parameters based on the calculated gradients during backpropagation and sets the step size for optimiser updates.
Sequence Length	512	Bert has a Maximum of 512.
Epoch	7	The algorithm iterates through the dataset, adjusts parameters, and improves performance.
Optimiser	Adam (Adaptive Moments)	An optimiser in machine learning updates the model's parameters to minimise the loss function. It adjusts based on gradients and ensures accurate predictions.

4.2. Performance Metrics

Performance metrics are the measurements used to assess the accuracy and quality of a model's predictions or classifications in machine learning, data analysis, and more. In this experiment, authors have evaluated the BERT-SPC using accuracy (Acc.) and F- measure (F1-score).

- As mentioned in equation 1. This is for evaluating accuracy. The accuracy of predictions is determined by the ratio of correctly predicted instances to the total number of instances in the dataset. This formula serves as a primary measure of overall correctness. Accuracy is calculated by counting the number of correct predictions where the model's predictions match the actual labels and dividing it by the total number of instances in the dataset used for evaluation. In conclusion, accuracy is an essential measurement in classification tasks [16].

$$A_{acc} = \frac{TP + TN}{TP + FP + FN + TN} \tag{1}$$

- As mentioned in equation 2, this is for evaluating Precision; when evaluating the predictive value, it is essential to consider the ratio of accurate optimistic predictions to the total predicted positives. This metric explicitly assesses the accuracy of optimistic predictions [16].

$$P_{\text{Prec}} = \frac{TP}{TP + FP} \tag{2}$$

- Equation 4 is used for evaluating Recall, the true positive rate and represents the number of accurate positive predictions to the total number of actual positives. Its main objective is to capture the maximum amount of true positive results. When assessing performance, this metric is crucial [16].

$$R_{Rec} = \frac{TP}{TP + FN} \tag{3}$$

- Equation 4 is used for evaluating F1 when evaluating performance; it's essential to consider both precision and recall. This gives a more accurate picture by factoring in false positives and negatives. In conclusion, conflicts can be better understood by using this balanced measure.

$$F1 = \frac{2 * P_{\text{Prec}} * R_{\text{Rec}}}{P_{\text{Prec}} + R_{\text{Rec}}} \tag{4}$$

5. Model Comparison and Results

We used Google Colab for our experimentation, which provided a powerful and efficient platform for our research. Table 3 compares the performance of three datasets: BERT-SPC, baseline, and state-of-the-art research. The proposed model outperformed the others in terms of performance.

- MemNet: The MemNet model [20] The model predicts sentiment using the last sentence representation and applying an attention mechanism to word embeddings multiple times.

- IAN: The LSTM network in the IAN model [21] extracts hidden states for target and context using word embeddings. An attention vector captures significant information, and a nonlinear layer produces the final classification outcome.

- ATAE-LSTM: To categorise emotional polarity, the LSTM [22] model first learns the coding of each feature before combining it with other aspects. This approach combines word representation and aspect embedding. This representation serves as the LSTM model's input, creating a hidden representation that holds the semantic details of the aspect. For modelling purposes, this model only considers one component at a time.

- AEN-BERT: This is a pre-trained attentional encoder network designed to solve the problem of aspect polarity classification. When used for SA based on aspect words, the model requires the input to be formatted as "[CLS]+ complete sentence + [SEP] + aspect word + [SEP]" [23].

- AEN-Glove: AEN comprises an output layer, a target attention layer, an attentional encoder layer, and GloVe/BERT embedding layers. There are two models: AEN-GloVe [**?**] and AEN-BERT.

- BERT-SPC: A pre-trained BERT model improves LCF-ATEPC's performance on sentence-pair classification for the APC subtask [24].

Table 3. Performance comparison of three different datasets

Models		**Twitter**		**Restaurant**		**Laptop**	
Baseline line Models	MemNet	66.30	65.90	78.16	65.83	70.33	64.09
	IAN	-	-	78.60	-	72.10	-
	ATAE-LSTM	-	-	77.20	-	72.10	-
	AEN-BERT	74.71	73.13	83.12	73.76	79.93	76.31
	AEN-Glove	72.83	69.81	80.98	72.14	73.51	69.04
	BERT-SPC	73.55	72.14	84.46	76.98	78.99	75.03
(Proposed)	**BERT-SPC (K=7)**	**74.71**	**72.65**	**84.98**	**76.98**	78.52	74. 21

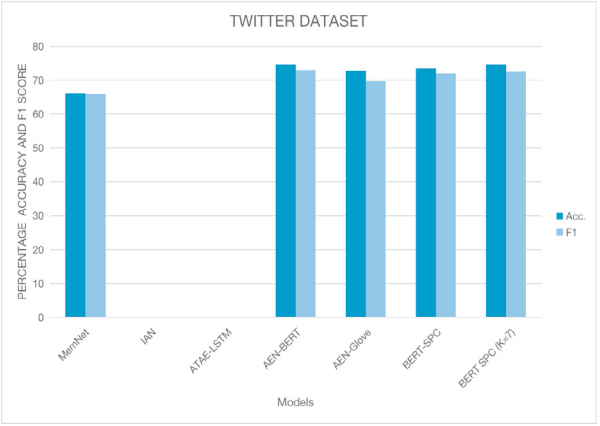


Fig. 6. Accuracy and F1 on Twitter Dataset

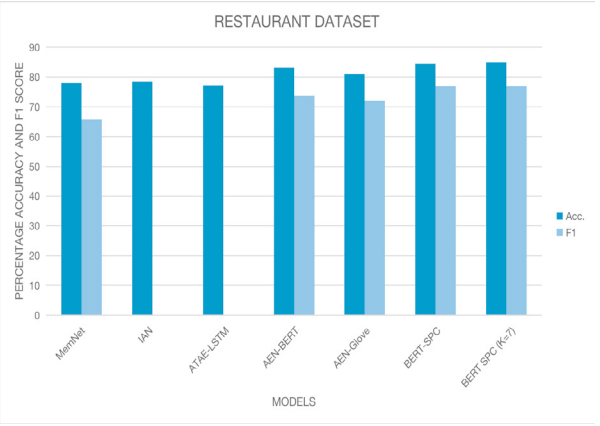


Fig. 7. Accuracy and F1 on Restaurant Dataset

Both baseline models and suggested models should be taken into account. Figure 4 shows an increase in F1 of 0.52% and accuracy of 1.16%. Figure 5 illustrates an improvement in accuracy of 0.52%. However, as can be seen from the accuracy findings of 74.71, 72.65, 84.98, 76.98, 78.52, and 74.21 in Figure 4, it should be highlighted that this model does not work well for laptops.

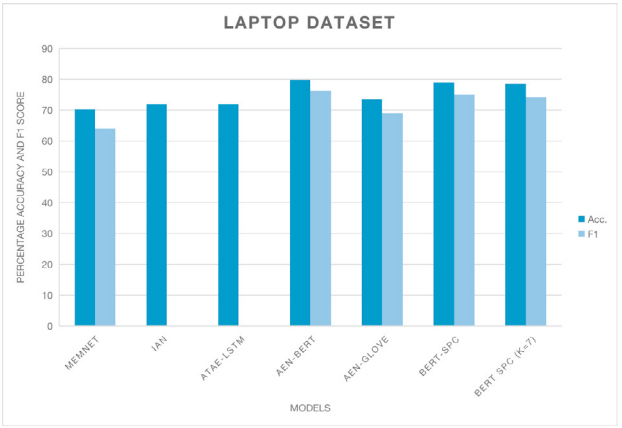


Fig. 8. Accuracy and F1 on Laptop Dataset.

6. Conclusion and Future work

This study examined baseline and suggested methodologies while investigating ABSA models. The study's findings demonstrated the proposed model's strong potential, with a noticeable accuracy and F1 score improvement. The suggested model can enhance sentiment analysis accuracy, as evidenced by the 1.16% increase in accuracy and the 0.52% rise in F1 score. The study did, however, also point out some of the model's shortcomings, particularly in terms of reliably gauging opinions towards computers. The accuracy scores ranged from 72.65% to 84.98%, highlighting the model's difficulties in that area. These findings highlight the model's limitations and offer potential directions for model improvement. It will be vital to incorporate these findings as the ABSA sector develops to increase the precision and applicability of sentiment analysis. In future work, authors plan to use this model for multimodal tasks involving both images and text. Authors aim to further improve the model's ability to process and analyze visual and textual information simultaneously, and to explore new applications in areas such as image captioning, visual question answering, and more. With the growing importance of multimodal data in various fields.

- Conflict of interest: The authors declare no conflict of interest regarding the publication of this manuscript.

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